SS-S-736D/GEN
June 1, 1989
SUPERCEDING
SS-S-736C
May 8, 1969

FEDERAL SPECIFICATION

STONES, SHARPENING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Fedearl agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers abrasive stones for the sharpening of edged tools and instruments. This specification may involve hazardous materials, and procedures and does not purport to address the safety or enviornmental problems associated with it's use. It is the responsibility of the user of this specification to consult and establish appropriate safety, health, and chemical disposal practices in accordance with all applicable state, local, and Federal regulations.

1.2 Types, classes, styles, and shapes (see SS-S-736/1 through SS-S-736/11).

Type I. Mounted-in-the-box stone.
Class B - Natural (Novaculite).
Style 1 - Retangular.

Type II. Unmounted stones.
Class A - 1 - Artificial (Aluminum oxide).
Class A - 2 - Artificial (Silicon carbide).

Style 1 - Retangular.
Style 2 - Half Round.
Style 3 - Round.
Style 4 - Square.
Style 5 - Triangular.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: General Services Administration, Industrial Engineering Group (7FXEI), 819 Taylor St., Fort Worth, TX 76102

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FSC 5345

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Style 6 - Knife Blade.
Style 7 - Point.
Style 9 - Silversmiths' oval tapered.
Style 12 - Reamer.
Style 13 - Gouge slip.
Style 14 - Round edge slip.

Class B - Natural (Novaculite).

Style 1 - Retangular.
Style 3 - Round.
Style 4 - Square.
Style 5 - Triangular.
Style 6 - Knife Blade.
Style 7 - Point.
Style 8 - Oval.
Style 14 - Round edge slip.

Type III. Taper Round Stones.
Class A - 1 - Artificial (Aluminum oxide).

Type IV. Scythe Stones.
Class A - 2 - Artificial (Silicon carbide).

Type V. Knife, Sharpening Stones.
Class A - 2 - Artificial (Silicon carbide).

Type VI. Grooved Surgical and Dental Stones.
Class B - Natural (Novaculite).

Type VII. Ax Sharpening Stones.
Class A - 2 - Artificial (Silicon carbide).

Type X. Shoemakers' Stones.
Class A - 2 - Artificial (Silicon carbide).

1.3 Part Number. Sharpening stones conforming with this specification shall be identified by a part number configuration consisting of identification of the general specification number, the applicable specification sheet (see paragraph 3.1), and the dash number which is an individual item identification number shown on the specification sheet. An example of the part number configuration is shown below. This part numbering system is intended for identification and cross-indexing of the item within the Federal cataloging system. Part numbers are not required to be placed on the product or container.

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S736/XX-XX
- S736 — General specification number.
- XX — Specification sheet number (see para. 3.1 & 6.3).
- XX — Dash number.

EXAMPLE: S736/1-Ø1

1.4 Dimensions. The dimensions of the stones shall be as shown in the applicable tables of the specification sheets (see 6.3).

1.5 Grits. Artificial abrasive stones shall be furnished in the grits described in section 3 for each class as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on the date of the invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material.
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-A-3816 - Abrasives and Abrasive Products, for Shipment and Storage, Packaging and Packing of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of the invitation for bids or request for proposal shall apply.

American National Standards Institute:

ANSI B74.18 - Specifications for Grading of Certain Abrasive Grain on Coated Abrasive Products.

ANSI B74.12 - Specifications for the Size of Abrasive Grain - Grinding Wheels, Polishing and General Industrial Uses.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references specified herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supercede applicable laws and regulations unless a specific exemption has been obtained.

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ANSI standards are available from the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets (see paragraph 6.3).

3.2 Material.

3.2.1 Artificial Stone, Class A-1 - Aluminum oxide and A-2 - Silicon Carbide.

3.2.1.1 Manufacturing process, classes A-1 and A-2. These classes of stones shall be made of the best electric furnace abrasives and suitable bonding materials, and shall be made in accordance with the best commercial practice under what is known commercially as the vitrified process (except type V cylindrical shaped stones). Type V cylindrical shaped stones (see figure 1-A of SS-S-736/8) shall be constructed with an internal reinforcing wire or rod to resist breakage and, unless otherwise specified (see 6.2), the stone shall be made by the sodium silicate bonding method.

3.2.2 Natural Stone, Class B.

3.2.2.1 Natural stones shall be of superior variety of novaculite, commonly known as Arkansas stone.

3.2.3 Grits.

3.2.3.1 Class A-1 artificial (aluminum oxide) and class A-2 artificial (silicon carbide). Classes A-1 and A-2 stones shall be furnished in extra-fine, fine, medium, coarse, extra-coarse and combination grits, as specified. Not less than 50 percent of the thickness of the combination stone shall be coarse grit and the remainder shall be fine grit; and not less than 42 percent of the stone shall be fine grit. The two grits shall be vitrified together. The minimum abrasive content of the stones shall be 65 percent and the maximum 85 percent. A combination of two classes of materials shall be bonded together either by the vitrified method or with a nonwater soluable, and oil resistant adhesive.

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3.2.3.1.1 Grit sizes for classes A-1 and A-2 shall be as follows:

(a)	Extra-fine:	400 to 500.
(b)	Fine:	280 to 320.
(c)	Medium:	150 to 240.
(d)	Coarse:	90 to 120.
(e)	Extra-coarse	60 to 80

3.2.3.1.3 Unless otherwise specified (see 6.2), class A-1 shall be oil-impregnated and class A-2 shall not be oil-impregnated.

3.2.3.2 Class B, natural (novaculite). Class B stones shall be furnished in hard (slow-cutting) and soft (medium cutting), as specified (see 6.2). Class B stones shall not be oil-impregnated.

3.2.3.2.1 Densities for class B, natural (novaculite) shall be as follows:

(a) Hard (slow cutting) - 2.30 g./cc, minimum.
(b) Soft (medium cutting) - 2.0 to 2.29 g./cc.

3.2.4 Type I stones. Type I stones shall be similar to figure 1 of SS-S-736/1, shall have a rectangular cross section, and mounted in a box. Unless otherwise specified, type I stones shall be finished on all sides except the lower surface which is cemented in the box. Edges of the stone may be slightly chamfered to prevent chipping. The stone shall be furnished firmly cemented in a mounting case with cover.. Material for the mounting case and cover shall be made of good commercial grade hardwood (species and grade at the manufacturer's option) or molded metal. The case shall incorporate a non-slip bottom surface.

3.3 Durability and utility. Sharpening stones shall possess sufficient strength to withstand normal service without chipping, breaking, or otherwise becomming damaged. They shall also show good sharpening qualities.

3.4 Identification marking. Each stone shall be prominently labeled or otherwise marked in a legible manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined. In addition, the stone or container shall be marked or labeled to indicate whether the stone should be oil-impregnated or not, and if it needs oil-impregnation, was it factory accomplished. Also, it should be stated that when using the stone, either no lubrication is required or the lubrication that shall be used will be oil, oil or water, or water only.

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3.5 Dimensional tolerances. Dimensional tolerances for class A-1, A-2, and B stones shall be in accordance with Table I unless otherwise specified (see 6.2).

TABLE I
Dimensional tolerance for sharpening stones

DIMENSIONS	
Inches	Tolerance
From 1/16 to 9/16	± 1/32
From 5/8 to 1-15/16	± 1/16
From 2 to 3-15/16	± 1/8
From 4 and greater	± 3/16

3.6 Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that the supplies and services conform to prescribed requiements.

4.2 Sampling for acceptance.

4.2.1 Inspection lot. For purposes of sampling, a lot shall consist of all stones of the same type, class, style, and size offered for delivery at one time. The sample unit shall be one stone.

4.2.2. Visual and dimensional examination. Sampling for this examination shall be in accordance with MIL-STD-105, inspection level S-2. The Acceptable Quality Levels (AQL's) for defects listed in Table II shall be 2.5 percent defective for major defects and 6.5 percent defective for total defects.

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4.2.3. Sampling for tests. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S-2. The Acceptable Quality Levels (AQL) for tests shall be 4.0 percent defective.

4.2.4 Sampling for examination of preparation for delivery. Sampling for this examination shall be in accordance with MIL-STD-105, inspection level S-2. The Acceptable Quality Level (AQL) shall be 6.5 expressed in terms of defects per hundred unit.

4.3 Quality Conformance Inspection.

4.3.1 Stones selected in accordance with 4.2.2 shall be examined for defects listed in Table II.

TABLE II
Classification of defects, end item

DEFECTS	MAJOR	MINOR
Type, class, style, or grit not as specified.	X	
Dimensions not within tolerance.		X
Breaks or chips.	X	
Combination types not bonded together as specified.	X	
Handles not securely fastened to stone, as applic.	X	
Handles not finished as specified, as applicable.		X
Type I Stones:		
Not secured in box.	X	
Wood cracked or split, as applicable.	X	
Marking not in accordance with 3.4.	X	
Other defects affecting function or serviceability.	X	
Other defects not affecting function or serviceability.		X

4.3.2 Tests. Each of the sample stones selected in accordance with 4.2.3 shall be subjected to the tests specified in 4.4.3. Unless otherwise specified (see 6.2) the test requirements in 4.4.1 and 4.4.2 shall be waived when the supplier furnishes acceptable proof of in-process control of abrasive content and of grit size. In case of dispute, all tests in 4.4 shall be conducted.

4.4 Test methods.

4.4.1 Test for determination of abrasive percentage. The test procedure for determination of abrasive percentage of class A-1 and A-2 stones shall be as follows:

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(a) Prepare and weigh a sample of a single class and grit size (fine, medium, or coarse) and place in a lead dish. Overspread with hydrofluoric acid, cover and let stand 24 hours. Pour off acid, break remains with stirring rod, wash by decantization, dry and weigh residuum. The abrasive percentage shall be calculated as follows:

$$\text{Abrasive Percentage} = \frac{B}{A} \times 100$$

A = Weight of sample
B = Weight of Residuum

4.4.2 Test for determination of grain size. To determine grain size, the grit obtained in 4.4.1 above shall be graded in accordance with ANSI B74.12 for grit sizes 240 and coarser. Grit sizes 280 and finer shall be graded in accordance with ANSI B74.18.

4.4.3 Test for determination of specific gravity of class A-1, A-2, or B stones. The test procedure for determination of the specific gravity of stones shall be as follows:

(a) Prepare a sample of a single class and grit size and weigh dry. Dip the stone in molten parafin wax, until completely coated, and weigh again. Reweigh coated sample under water. Calculate the specific gravity of the sample stone as follows:

$$\text{Specific Gravity} = \frac{A}{B - \left(\frac{B - A}{S} + C \right)}$$

A = Weight of Sample
B = Weight of Coated Sample
C = Weight of Submerged Sample
S = Specific Gravity of Paraffin

4.5 Examination for preparation for delivery. An examination of sample unit, intermediate, or shipping containers (as applicable), selected in accordance with 4.2.4 shall be made to determine that the packaging, packing, and marking comply with requirements of section 5 and as specified in the contract or purchase order.

5. PREPARATION FOR DELIVERY

5.1 Unit packaging. Unit packaging shall be level A, B, C, or as otherwise specified in the contract or purchase order (see 6.2).

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5.1.1 Level A. Stones shall be packaged in accordance with the level A requirements of MIL-P-3816.

5.1.2 Level B. The number of stones as specified in the contract or purchase order, shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-636, PPP-B-665, or PPP-B-676. The box shall be closed and secured to prevent accidental opening.

5.1.3 Level C. Stones shall be packaged in accordance with the supplier's normal commercial practice.

5.2 Intermediate packaging. The number of unit packages as specified in the contract or purchase order, shall be intermediate packaged in a close-fitting box conforming to PPP-B-566, PPP-B-636, PPP-B-665, or PPP-B-676 or as otherwise specified in the contract or purchase order. The box shall be closed and secured to prevent accidental opening.

5.3 Packing. Packing shall be level A, B, C, or as otherwise specified in the contract or purchase order (see 6.2).

5.3.1 Level A. The unit packages in quantities as specified shall be packed in accordance with the level A requirements of MIL-P-3816.

5.3.2 Level B. The number of unit or intermediate packages as specified in the contract or purchase order, shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with appendix to PPP-B-636.

5.4 Marking.

5.4.1 Civil agencies. In addition to any special markings specified in the contract or purchase order, packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military activities. In addition to any special markings specified in the contract or purchase order, packages and shipping containers shall be marked in accordance with MIL-STD-129 and MIL-P-3816 as applicable.

6. NOTES

6.1 Intended use. Sharpening stones covered by this specification are intended for use in sharpening the cutting edges of edged cutting tools. For the variety of tools for which these stones are designed, see 1.2.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification and the appropriate specification sheet (see 6.3).
(b) Type, class, style, size and grit required (see 1.2 thru 1.5 and 3.1 thru 3.5).
(c) Oil-impregnation of class A-2, if required (see 3.2.3.1.3).
(d) If dimensional tolerances are different (see 3.5).
(e) If 17-A knife sharpening stone is to be made by other than the sodium silicate bonding method (see SS-S-736/).
(f) If lot-by-lot testing is required in lieu of in-process controls (see 4.3.2).
(g) Levels of preservation, packaging, and packing desired (see 5.1 thru 5.3).
(h) Special marking if required (see 5.4.1 and 5.4.2).

6.3 Specification sheets. Refer to table III which indicates the specification sheets applicable to the various types, classes and styles.

Table III. Classification and dimensional tables

Specification sheet	Type	Class	Styles or Shapes
SS-S-736/1	I	B	1
SS-S-736/2	II	A-1	1, 2, 3, 4, 5, 6, 7, 9, 12, 13, 14
SS-S-736/3	II	A-2	1, 2, 4, 5, 6, 14
SS-S-736/4	II	B	1, 3, 4, 5, 6, 7, 8, 14
SS-S-736/5	II	A-1/A-2	1 (COMBINATION)
		B/A-1	1 (COMBINATION)
SS-S-736/6	III	A-1	
SS-S-736/7	IV	A-2	
SS-S-736/8	V	A-2	
SS-S-736/9	VI	B	
SS-S-736/10	VII	A-2	
SS-S-736/11	X	A-2	

6.4 Cross reference data. The types, classes, grits, styles, and sizes in Federal Specification SS-S-736C directly correspond to the types, classes, grits, styles and sizes in this document and associated specification sheets. Some types and classes shown in SS-S-736C have become obsolete for government use and are not covered by this revision.

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6.5 The illustrations depicted in the applicable specification sheets are descriptive rather than restrictive, and are not intended to preclude the purchase of sharpening stones otherwise conforming to this specification.

MILITARY INTERESTS:

Review activities:

Air Force - 84
Army - AV

User activity:

Navy - YD

PREPARING ACTIVITY:

GSA - FSS (7FXE)

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